Geodesic: a parallel test-particle geodesic tracker for the Einstein Toolkit

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Abstract

We present `Geodesic`, a new thorn for the Einstein Toolkit that integrates test particles on geodesics of a Kerr spacetime described in Kerr–Schild Cartesian coordinates. The thorn provides two modes of metric evaluation: an exact analytic Kerr–Schild metric, which is independent of the computational grid and whose vacuum character ($R_{\mu\nu} = 0$) is verified symbolically, and three grid-metric samplers selected by a single parameter: the default 27-point local Lagrange interpolant (a polynomial of degree at most two in each coordinate), a fourth-order pointwise Lagrange interpolant built on the `AEILocalInterp` thorn, and a second-order pointwise Lagrange interpolant (historically named after the `CarpetInterp` driver). In all three, the contribution of each particle is computed entirely independently of all others. This independence makes the particle loop embarrassingly parallel; with an OpenMP `parallel for` over the particles and no global interpolation or inter-particle communication, the thorn scales to 10^4 particles on a multicore node: with 10^4 particles on 88 threads, one full step (particle integration plus the surrounding framework) takes $\sim 0.15\,\mathrm{s}$ of wall time, about $15\,\mu\mathrm{s}$ per particle per step; embedded in a full `GRHydro` evolution, the integration of 10^4 particles accounts for only $\sim$3.1% of the total wall time, measured while the node was shared with three other concurrent instances ($\sim$22 threads each). The geodesic equation is integrated by an adaptive Runge–Kutta–Fehlberg (RKF45) scheme with a per-step renormalization of the four-velocity. We verify the integrator by (i) tracking three bound orbits (circular, tilted, eccentric) for $T = 1000\,M$, where the energy E, the axial angular momentum L_z and the norm $g_{\mu\nu}u^\mu u^\nu = -1$ are conserved to $\sim 10^{-9}$; (ii) demonstrating grid convergence of the interpolation mode with a measured local order of 3.7 (formal order three) against an independent reference integration of the exact metric; and (iii) comparing the three grid samplers on a precessing bound orbit of 100 particles: no particle is lost to the excision sphere over $\sim$ 2.5 radial periods, and the maximum drift in E and L_z is $(4.1 \times 10^{-5}, 8.3 \times 10^{-4})$, $(2.6 \times 10^{-4}, 1.4 \times 10^{-2})$ and $(8.4 \times 10^{-5}, 7.7 \times 10^{-4})$ for the fourth-order, second-order and default samplers, respectively (Sec. 5.4). The three grid samplers are further verified on a 3-level AMR mesh, where the same orbit is reproduced to within 10^{-2} of the unigrid reference (Sec. 5.5). A strong-scaling test with 10^4 particles reaches a speedup of 9.2× (efficiency 10.4%) on the 88 logical threads of a two-socket, 44-core hyperthreaded node. The thorn is released under the GPLv2+ license together with the configuration files used in this work.

1 Introduction

Test-particle geodesics are a basic diagnostic in numerical relativity and in modelling of accretion flows around black holes [Poisson, 2004]: they provide reference orbits against which self-consistent

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hydrodynamic and electromagnetic solutions can be compared [Blandford and Znajek, 1977], they define the orbital structure used to interpret line profiles and disk precession [Bardeen, 1973, Gralla and Lupsasca, 2020], and they serve as a validation tool for the spacetime produced by a simulation [Kerr, 1963, Bardeen et al., 1972, Wilkins, 1972]. The standard frameworks for general-relativistic simulations of compact objects, the Cactus code framework [Cactus Code, 2024] and its extension, the Einstein Toolkit [Löffler et al., 2012], do not ship a geodesic tracker for test particles in Kerr–Schild coordinates, which is the coordinate system used by most Kerr initial-data and spacetime-evolution thorns. Users who need geodesics today must either write their own integrator or post-process the evolved metric fields, and in both cases the cost of a global interpolation of the metric at every particle position prevents the straightforward parallel scaling that the problem structure allows.

`Geodesic` is a thorn for the Einstein Toolkit that supplies this functionality. Its design goals are (i) to support both an exact analytic Kerr–Schild metric and a metric interpolated from the simulation grid, so that the same particle set can be tracked on the analytic background or on an evolved spacetime; (ii) to keep the per-particle work completely local, so that the particle loop can be parallelized with OpenMP without any inter-particle communication; and (iii) to be a clean, auditable reference implementation of the geodesic equation in Kerr–Schild Cartesian coordinates, verified against an independent implementation.

The main contributions of this paper are:

1. a thorn that integrates the geodesic equation in Kerr–Schild Cartesian coordinates with an adaptive RKF45 scheme and a per-step four-velocity renormalization;
2. an exact analytic Kerr–Schild metric whose vacuum character is verified symbolically (Sec. 3.1);
3. three grid-metric samplers selected by one parameter — the default 27-point local Lagrange interpolant (degree at most two in each coordinate), a fourth-order pointwise interpolant built on the `AEILocalInterp` thorn, and a second-order pointwise Lagrange interpolant (historically named after the `CarpetInterp` driver) — in which every particle is an independent unit of work, validated by a strong-scaling run with 10^4 particles (Secs. 3.2–6);
4. a verification suite: conservation of E, L_z and the norm over $T = 1000\,M$ for three distinct bound orbits, a grid-convergence study showing a measured local convergence order of 3.7 (formal order three) for the default sampler, and a four-way comparison of the exact metric against the three samplers on a precessing orbit of 100 particles (Sec. 5).

The remainder of the paper is organized as follows: Sec. 2 describes the geodesic equation in the 3+1 formulation and the initialization of the particle four-velocity; Sec. 3 the exact mode and the three grid-metric samplers; Sec. 4 the implementation, including the OpenMP parallelization and the particle-lifecycle management; Sec. 5 the verification results; Sec. 6 the strong-scaling results; and Sec. 7 concludes.

2 Method

2.1 Geodesic equation

We work in Kerr–Schild Cartesian coordinates (t, x, y, z) with spin axis along $+z$, in which the Kerr metric of mass M and dimensionless spin a takes the form [Kerr and Schild, 1965]

$$g_{\mu\nu} = \eta_{\mu\nu} + \frac{2Mr^3}{D}\, l_\mu l_\nu, \qquad l_\mu = \left(1,\, \frac{ay + rx}{Q},\, \frac{ry - ax}{Q},\, \frac{z}{r}\right), \tag{1}$$

with $\eta_{\mu\nu} = \mathrm{diag}(-1,1,1,1)$, $Q = r^2 + a^2$, $D = r^4 + a^2 z^2$, and r the Kerr radial coordinate defined by $r^2 = (x^2 + y^2 + z^2 - a^2 + \sqrt{(x^2 + y^2 + z^2 - a^2)^2 + 4a^2 z^2})/2$. The vector l_μ is null, $g_{\mu\nu} l^\mu l^\nu = 0$, and the metric is a vacuum solution, $R_{\mu\nu} = 0$ (Sec. 3.1).

The four-velocity $u^\mu = \mathrm{d}x^\mu/\mathrm{d}\tau$ of a massive test particle obeys the geodesic equation

$$\frac{\mathrm{d}u^\mu}{\mathrm{d}\tau} = -\Gamma^\mu_{\alpha\beta}\, u^\alpha u^\beta, \qquad g_{\mu\nu} u^\mu u^\nu = -1, \tag{2}$$

where $\Gamma^\mu_{\alpha\beta}$ are the Christoffel symbols of $g_{\mu\nu}$. Because the metric is stationary and axisymmetric, two quantities are conserved along the geodesic: the specific energy $E = -u_t$ (Killing vector ∂_t) and the axial angular momentum $L_z = -y\, u_x + x\, u_y$ (Killing vector $-y\partial_x + x\partial_y$). Both are used as verification monitors.

2.2 Integration in proper time

The thorn integrates Eq. (2) in proper time τ with an adaptive fifth-order Runge–Kutta–Fehlberg scheme (RKF45, Fehlberg 1969), with an absolute error tolerance of 10^{-8} and zero relative tolerance. Each Cactus time step advances the particle until its coordinate time reaches $t+\Delta t$: the proper-time increment is predicted so that t lands on the target, and any overshoot is removed by a linear back-interpolation, so that all particles share the Cactus time axis. At the end of every Cactus time step the thorn renormalizes the four-velocity so that $g_{\mu\nu} u^\mu u^\nu = -1$ holds to roundoff, which removes the secular drift of the norm that pure adaptive integrators accumulate. The spatial components u^i produced by the ODE step are kept as they are, and the constraint is solved for u^t as a quadratic,

$$g_{00}\,(u^t)^2 + 2\, g_{0i}\, u^i\, u^t + \left(1 + \gamma_{ij} u^i u^j\right) = 0, \tag{3}$$

selecting the future-directed root. Outside the ergosphere ($g_{00} < 0$) the two roots have opposite signs and the unique positive root is unambiguous; inside the ergosphere the same u^i can in fact correspond to two future-directed solutions (a geometric effect of frame dragging), and the root is chosen by continuity, i.e. the one closest to the u^t produced by the ODE step before renormalization.

2.3 Initialization of the four-velocity

The particle initial conditions are specified in the par file as a position (x, y, z) and the contravariant four-velocity components $u^\mu = (u^t, u^x, u^y, u^z)$ in Kerr–Schild coordinates. At the first time step the thorn evaluates the metric at the initial position and checks the normalization $g_{\mu\nu} u^\mu u^\nu = -1$. If it holds to a tolerance of 10^{-6}, the input is used unchanged and the log says so; otherwise u^t is re-solved from Eq. (3) with u^i held fixed, and the correction (before/after values) is written to the log. The check is therefore idempotent: feeding the values logged at initialization back into the par file changes nothing. The 10^{-6} tolerance is loose on purpose; the per-step renormalization described above restores the constraint to roundoff on the first step regardless. All orbits used in this paper are generated by first integrating the exact-metric geodesic equation with an independent Python implementation (RK4, $\Delta t = 0.02\, M$) and then writing the resulting exact four-velocity directly into the par file; the initial data are listed in Table 1.

3 Metric evaluation

3.1 Exact Kerr–Schild mode

With `Exact = yes` the thorn evaluates the Kerr–Schild metric and its coordinate derivatives analytically at the particle position (Eq. (1)), so that the result is independent of the computational

grid. The analytic Christoffel symbols are obtained from the analytic first derivatives of the metric; no finite differences enter the evolution. As a self-consistency check we verify the vacuum character of the implemented metric symbolically with SymPy: the components of the thorn's metric are checked term-by-term against the Kerr–Schild form Eq. (1) (which also proves l_μ is null), and the Ricci tensor $R_{\mu\nu}$ is assembled from the analytic metric and its analytic inverse $g^{\mu\nu} = \eta^{\mu\nu} - H\, l^\mu l^\nu$ and evaluated at five generic points to 40-digit precision, giving $\max|R_{\mu\nu}| < 10^{-170}$, i.e. the implemented metric is a vacuum solution to beyond machine precision.

3.2 Grid-metric samplers

With `Exact = no` the metric is taken from the simulation grid: the thorn reads the `ADMBase` lapse, shift and spatial metric, assembles the covariant metric $g_{\mu\nu}$ at every grid point, and obtains the metric derivatives there with sixth-order central finite differences. For time-dependent grids the $\partial_t g_{\mu\nu}$ terms of Γ^μ_{00} and $\Gamma^\mu_{0\nu}$ are needed: the thorn computes them with a second-order backward finite difference, $(3L_0 - 4L_1 + L_2)/(2\,\Delta t)$, over the three most recent `ADMBase` time levels. A stationary metric is detected automatically (the two oldest levels agree to relative 10^{-12}), in which case $\partial_t g_{\mu\nu} = 0$ exactly and the difference is skipped. Within one particle step the connection is frozen at the grid's current time, so for a time-dependent metric the particle step is capped at the grid time step. How the metric and its first spatial derivatives are sampled at the (generally off-grid) particle position is selected by the parameter `metric_interp`; the three choices are described below. All three samplers are pointwise and local: the computation for one particle reads only the grid cells around it and writes no data used by any other particle. This is in contrast to a global (e.g. radial or tree-based) interpolation, which requires a collective pass over all particles and prevents independent parallel scaling.

local27 (default). For a given particle the thorn identifies the 27 grid points of the 3×3×3 cell stencil surrounding it and, for each metric component and each Christoffel symbol, fits the unique polynomial that is at most quadratic in each coordinate and interpolates the 27 stencil values — a 27-term tensor-product Lagrange interpolant [Berrut and Trefethen, 2004]. The fit is a single matrix–vector product with a precomputed 27×27 inverse Vandermonde matrix, and fields that are constant on the stencil collapse to their constant value without the product. The fit is then cached per particle: it is built once per Cactus time step (and re-evaluated within the step only if the particle crosses into an adjacent stencil cell), and every subsequent adaptive RKF45 substep and stage evaluates the cached polynomials alone — within a step the right-hand side touches neither the grid nor the finite differences. The expensive part (sampling the metric and Christoffel symbols on the 27 stencil points and forming the fits) is thus amortized over the whole step, and the per-stage cost of the integration reduces to a small number of polynomial evaluations. For smooth fields this interpolant is locally third-order accurate, $\mathcal{O}(\Delta x^3)$, while the finite-difference error of the stencil values is $\mathcal{O}(\Delta x^6)$ and does not limit the convergence. The sampling points are clipped to the cells interior to the local processor box (at least five cells from the local boundary), so the stencil never reaches the ghost region and the sampler is safe under MPI domain decomposition.

aei (fourth-order pointwise). `metric_interp = aei` evaluates, at the particle position, a fourth-order pointwise Lagrange interpolant of the ten 3+1 fields (lapse, three shifts, six spatial-metric components) and of their first spatial derivatives, using the `AEILocalInterp` thorn [Einstein Toolkit, 2026a]; the metric and its Christoffel symbols are then composed pointwise from these sampled values. Unlike `local27`, no per-particle fit is cached: the interpolant is re-evaluated at every RKF45

stage. The fourth-order stencil reaches two grid points beyond the local box in each direction, so in a multi-processor run it can extend into the ghost region, whose values the surrounding framework does not synchronise at every step; particles that ride on a processor plane (as in our four-processor test, where all 100 particles move in the $z = 0$ plane) then sample stale or uninitialised ghost data, and the run diverges from the single-processor reference within a few grid steps (Sec. 5.4). The official remedy is the `AEILocalInterp` parameter `N_boundary_points_omit[]`, which shrinks the active domain by the stencil width so that the stencil never crosses a processor boundary; we recommend single-processor (OpenMP-only) runs, or that parameter, for `aei`.

carpet (second-order pointwise). `metric_interp = carpet` evaluates, at the particle position, a second-order (quadratic) pointwise Lagrange interpolant of the ten 3+1 fields: a tensor-product stencil of three nodes in each coordinate direction, with the spatial derivatives taken as the analytic derivatives of the same interpolant (the same order as the seven-point centred differences used by the original `CarpetInterp` driver [Einstein Toolkit, 2026b]). The metric and its Christoffel symbols are then composed pointwise from the sampled values, as for `aei`. The stencil values are read from the thread-local copy of the refinement level that contains the particle, so the sampler is thread-safe and works on every rank in a multi-processor run; the historical name is retained. As with `aei`, no per-particle fit is cached: the interpolant is re-evaluated at every RKF45 stage.

Temporal interpolation. With `time_interp = off` (the default) the metric is read from the newest available `ADMBase` time level and $\partial_t g_{\mu\nu}$ is supplied by the three-level backward difference described above. With `time_interp = linear` the two most recent levels are blended linearly with the fractional part of the particle time within the current grid step, $\theta = (t - t_{\rm bg})/\Delta t$: for `local27` the two levels are fitted separately and the polynomials are blended, and for `aei` the interpolated values and derivatives are blended. The linear option is not supported by `carpet`.

4 Implementation

`Geodesic` is written in C as a Cactus/Einstein Toolkit thorn (5681 lines in the main integration file). The particle state $(t, x, y, z, u^t, u^x, u^y, u^z)$ is stored in Cactus interface arrays `particle_tt`, `particle_tx`..., and the integration loop is a plain C loop over the particles, wrapped in an OpenMP `parallel for` with `schedule(dynamic,1)`:

```
#pragma omp parallel for schedule(dynamic,1)
for (mi = 0; mi < particle_n_total; mi++) {
    /* per-particle RKF45 step on the locally evaluated metric */
}
```

There is no shared state inside the loop: each thread evaluates the metric for its own particle (either analytically, Sec. 3.1, or from the grid with one of the local samplers of Sec. 3.2) and advances its own Runge–Kutta state. The only synchronization is the OpenMP barrier at the end of the parallel region, once per Cactus time step.

Particle lifecycle. What happens to a particle that leaves the computational domain or whose four-velocity fails the normalization check (no future-directed root of Eq. (3)) is selected at run time with `particle_loss_policy`: `drop` (the default) marks the particle state as NaN and stops tracking it, `flag` freezes it at its last valid state, and `respawn` re-places it at a uniformly random

position inside the safe box, the grid domain inset by seven cells, subject to the rejections $r >$ excised_radius $+ 1.5$ (outside the excision sphere) and $r <$ initial_radius (inside the initial particle shell), with $u^i = 0$ and a u^t placeholder; the per-step re-solve of Eq. (3) then normalizes the four-velocity on the next step. Every loss is reported to the log with the particle index. The scaling run of Sec. 6 uses `respawn`, which keeps the particle count constant over long runs and prevents a single lost particle from poisoning the statistics.

Diagnostics. With `particle_dump_every = k > 0` the thorn writes the full particle state $(t, x, y, z, u^t, u^x, u^y, u^z, \tau)$ to `geodesic_particles.txt`: the file begins with the initial state ($t = 0$) recorded before the first integration step, followed by one block of rows every k iterations (rank 0 only); this file is the data source for all verification figures below. `rverbose = yes` prints the RKF45 sub-step history to the log for debugging.

5 Verification

5.1 Test orbits

All verification runs use $M = 2$, $a = 0.5$ (the values of the `KerrSchild` initial data [Moreno et al., 2002]) and a 3+1 grid with Cartesian cells of size Δx. Three bound orbits are used (Table 1), each generated and validated independently in Python before the thorn run:

Table 1: Test orbits used for verification ($M = 2$, $a = 0.5$). u^t and u^x, u^y, u^z are the four-velocity components in the par file. $\Delta E/E$ is the value measured by the independent Python reference over the same interval.

orbit	initial (x, y, z)	u^t	u^x, u^y, u^z	radial range	ref. $\Delta E/E$
P0 circular ($r = 10$)	$(10, 0, 0)$	1.53428	(0, 0.67238, 0)	$r = 10$	1.2×10^{-9}
P1 tilted	$(-13.448, 1.180, 0)$	1.32221	$(-0.01827, -0.50381, 0.1)$	[12.86, 16.88]	5.0×10^{-9}
P2 eccentric	$(15, 0, 0)$	1.27607	(0, 0.45176, 0)	[12.03, 15]	5.3×10^{-9}

P0 is a circular orbit of radius $r = 10$ (azimuthal period $T_\phi = 143.4\,M$); P2 is an eccentric orbit with apocenter $r_a = 15$ and pericenter $r_p = 12.03$; P1 is P2's orbit given a small vertical kick at one point, tilting it out of the equatorial plane. Figure 1 shows the three orbits as integrated by the thorn (Exact mode, $T = 1000\,M$): P0 closes on itself in the equatorial plane, while the rosette patterns of P1 and P2 reflect the relativistic periastron advance of the Kerr spacetime.

5.2 Conservation of E, L_z and the norm

We integrate the three orbits simultaneously with `Exact = yes` for $T = 1000\,M$ (10^4 Cactus steps of $\Delta t = 0.1\,M$, i.e. about seven azimuthal periods of P0), on a 100^3 grid. Figure 2 shows the relative drift of E and L_z and the norm residual $|g_{\mu\nu}u^\mu u^\nu + 1|$ for all three orbits. The relative energy drift stays at the level of the independent reference integration ($\leq 3.0 \times 10^{-10}$) and the norm is held at the roundoff level ($\leq 8.9 \times 10^{-16}$) throughout, confirming that the RKF45 integrator with per-step renormalization reproduces the geodesic equation accurately over many orbital periods.

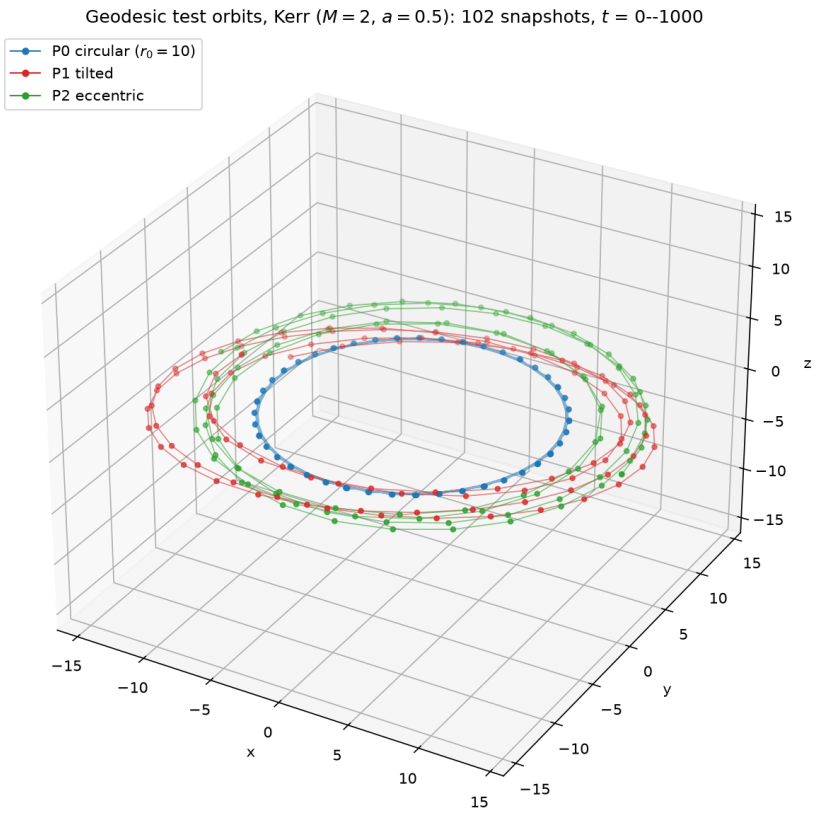


Figure 1: The three test orbits of Table 1 in the Kerr spacetime ($M = 2$, $a = 0.5$), integrated by the thorn in Exact mode over $T = 1000\,M$ (102 snapshots). P0, a circular orbit of radius $r = 10$ in the equatorial plane (blue); P1, P2's orbit given a small vertical kick, precessing out of the plane (red); P2, an eccentric orbit with $r_a = 15$, $r_p = 12.03$ (green).

5.3 Grid convergence of the interpolation mode

To quantify the accuracy of the 27-point interpolation mode, we track the circular orbit P0 for one full azimuthal period ($t = 150.1\,M$) on grids of 50^3, 100^3 and 200^3 cells ($\Delta x = 1.0,\ 0.5,\ 0.25$) with `Exact = no`, and compare the final particle position with an independent Python RK4 integration of the exact metric (step $0.02\,M$). Figure 3 shows the orbit traces and the final-position error versus grid spacing: the error decreases from 3.7×10^{-1} to 2.9×10^{-2} and 2.3×10^{-3} on the two refinements (factors of 12.9 and 12.8), a measured local convergence order of 3.7, and the exact-metric run agrees with the reference to 1.7×10^{-6}, which is the level at which two independent integrators differ. The measured order exceeds the formal third order of the interpolant: the formal order is a worst-case bound for smooth fields, and the realized error coefficient along this orbit is smaller than the bound would suggest; we report the measured rate and do not claim that the formal order is saturated.

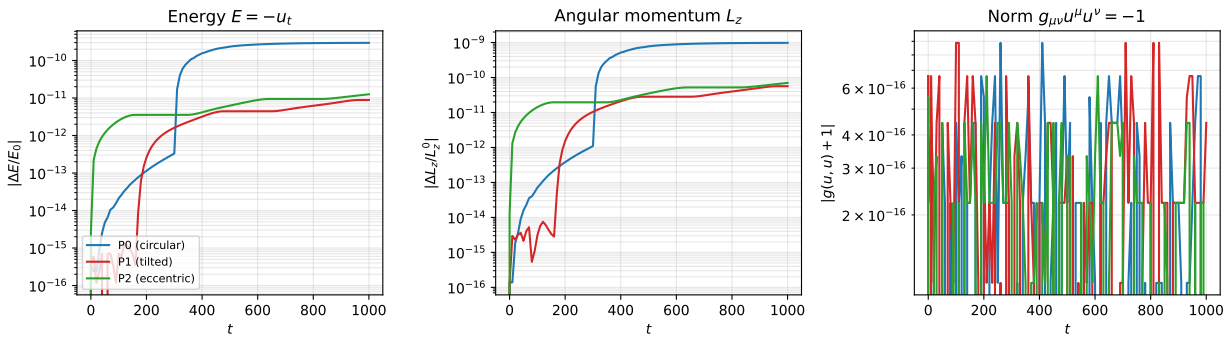


Figure 2: Conservation over $T = 1000\,M$ (Exact mode): relative drift of the energy E (left) and axial angular momentum L_z (middle), and the norm residual $|g_{\mu\nu}u^\mu u^\nu + 1|$ (right), for the three orbits of Table 1. All quantities are conserved at the $\lesssim 10^{-9}$ level.

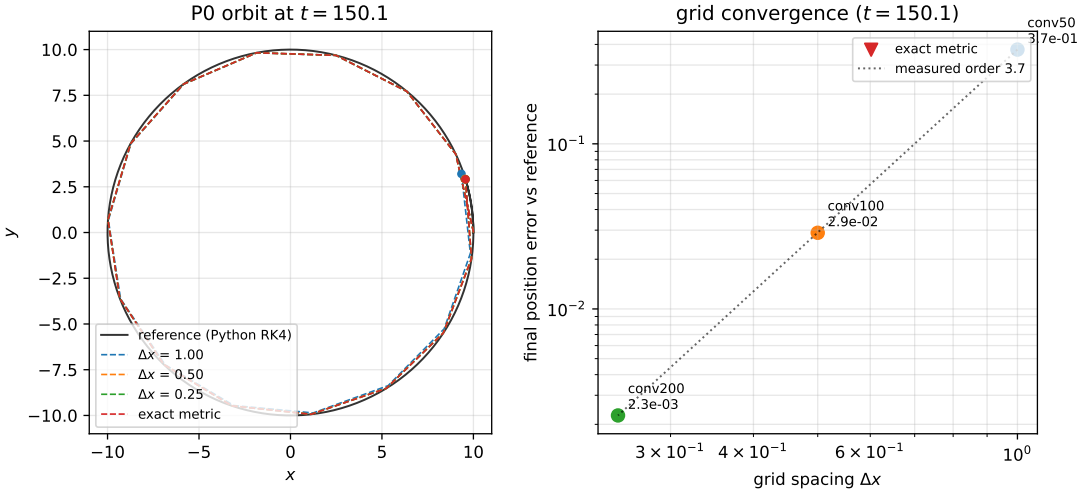


Figure 3: Grid convergence of the P0 circular orbit over one period ($t = 150.1\,M$). Left: orbit traces in the x–y plane for the interpolation runs ($\Delta x = 1.0, 0.5, 0.25$) and the exact-metric run, compared with the independent Python reference (solid). Right: final-position error versus grid spacing (markers), with a reference line of the measured order 3.7 (dotted).

5.4 Comparison of the grid samplers on a precessing orbit

To compare the three grid samplers on an orbit closer to production use, we track 100 particles that all move on the same bound equatorial Kerr ellipse ($M = 2$, $a = 0.5$, pericentre $r_p = 8.5$, apocentre $r_a = 64$, $E = 0.97434$, $L_z = 7.16068$), spread over the orbit, on the static Kerr–Schild field written onto a 144^3 base grid ($\Delta x = 1$) with three nested refinement boxes ($\Delta x = 0.5$, 0.25, 0.125 out to radii 36, 18, 9). The ellipse precesses by roughly 4π per radial period, so each particle sweeps the whole grid domain during the run, $t = 0 \ldots 3200\,M$ (about 2.5 radial periods); the particle state is dumped every 100 time units, and a particle is counted as lost when it crosses the excision sphere $r < 4$. The four runs are `Exact = yes` (reference, single process, 20 OpenMP threads), `metric_interp = aei` (single process, 20 threads), `carpet` (single process, 16 threads)

and `local27` (four processes, 16 threads each), the last being the multi-processor configuration of Sec. 6.

Table 2: Comparison of the three grid samplers on the precessing ellipse of Sec. 5.4 (100 particles, $t = 0 \ldots 3200\,M$; no particle lost in any run). ΔE, ΔL_z: maximum and mean absolute deviation of the conserved quantities from their initial values, over the run. "p0": state of particle 0 at the end of the run.

metric evaluation	run	lost	survivors	$\max\lvert\Delta E\rvert$ / mean	$\max\lvert\Delta L_z\rvert$ / mean	p0
exact (reference)	1 process, 20 threads	0	100	—	—	$r = 8.63$ (alive)
aei (order 4)	1 process, 20 threads	0	100	4.06×10^{-5} / 9.98×10^{-6}	8.32×10^{-4} / 3.13×10^{-4}	$r = 8.65$ (alive)
carpet (order 2)	1 process, 16 threads	0	100	2.63×10^{-4} / 5.00×10^{-5}	1.35×10^{-2} / 5.38×10^{-3}	$r = 8.69$ (alive)
local27 (default)	4 processes	0	100	8.38×10^{-5} / 1.69×10^{-5}	7.69×10^{-4} / 2.15×10^{-4}	$r = 8.69$ (alive)

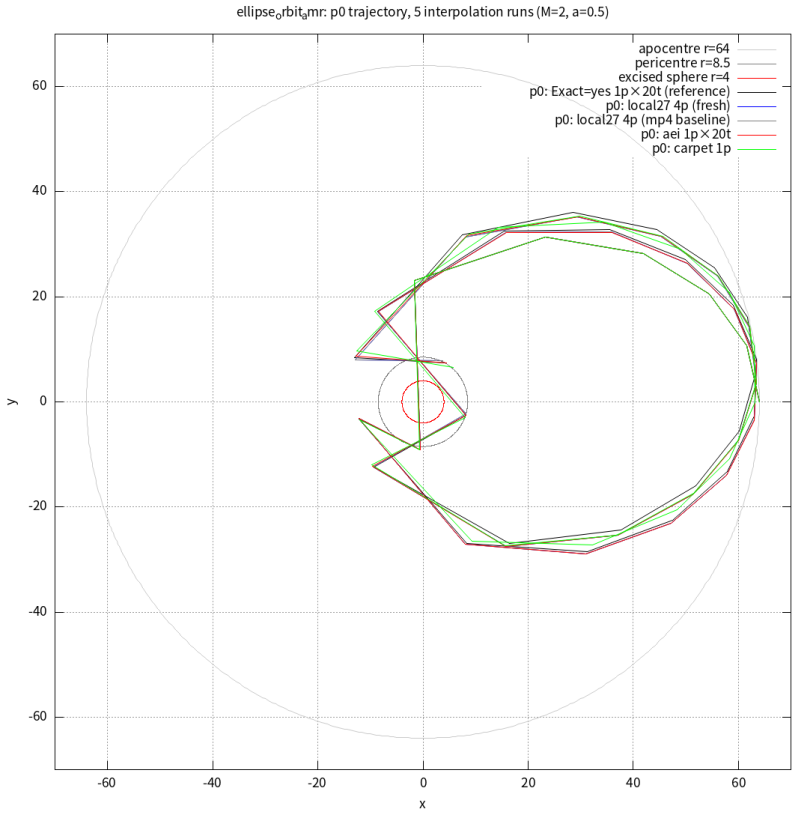


Figure 4: Particle 0 of the precessing-ellipse comparison (Sec. 5.4) in the x–y plane: exact reference (black), `local27` four-process run (blue), `aei` (red) and `carpet` (green). The dashed gray line is the earlier multi-processor `local27` baseline, bit-identical to the solid blue line. Circles: apocentre ($r = 64$), pericentre ($r = 8.5$) and the excision sphere ($r = 4$, dashed).

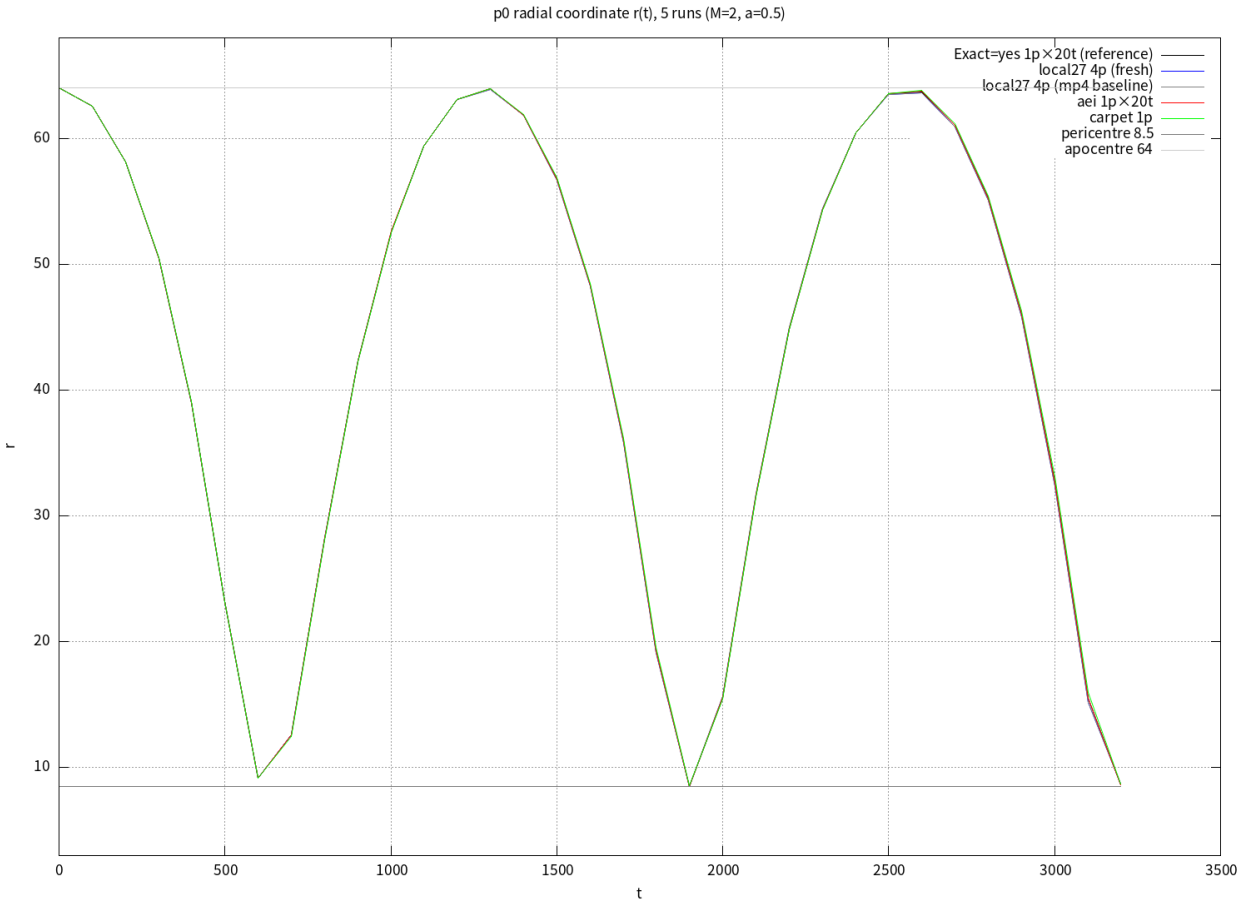


Figure 5: Radial coordinate $r(t)$ of particle 0 for the runs of Table 2. Dashed: pericentre ($r = 8.5$) and apocentre ($r = 64$) of the reference ellipse.

No particle is lost to the excision sphere in any run. The conserved-quantity drifts order as `aei` < `local27` < `carpet` (Table 2). This is not a ranking by formal order — `local27` is formally third order, higher than `carpet`'s second order — because the dominant error in the run is the $\sim 10^{-4}$ coherent discretization error of the grid itself (an exact Kerr–Schild field written onto a finite grid), which dwarfs the interpolation error of all three samplers, and what distinguishes them is how each treats that common grid error. The fourth-order pointwise `aei` sampler smooths it best; the second-order pointwise `carpet` sampler smooths it least; and `local27`, which re-fits its 27-point polynomial at the particle position on every RKF45 substep, reproduces the grid's coherent error rather than smoothing it, so along this orbit it effectively resolves the grid rather than the interpolant's higher-order content. This has no adverse consequence in production use: in a full `GRHydro` evolution or any grid-carried spacetime the grid field is the true metric, and there `local27`'s formally third-order interpolation is strictly better than `carpet`'s second order; `aei` is the choice that most faithfully reproduces the exact analytic background in a benchmark.

Run notes. The `local27` four-process run reproduces the earlier multi-processor baseline bit-for-bit (the particle dumps are identical), so the `local27` row of Table 2 is also the configuration used in Sec. 6. `aei` was run with a single process: in a four-processor run all 100 particles move in the $z = 0$ plane that coincides with a processor boundary, the fourth-order stencil extends

into the unsynchronised ghost region, and the run diverges from the single-processor reference (maximum position difference 1.4×10^{-3} at $t = 100$, 14.1 at $t = 200$) — the multi-processor limitation discussed in Sec. 3.2. `carpet` was run with a single process; its original implementation called the collective `CarpetInterp` driver and was single-processor by construction, whereas the current pointwise implementation is thread-safe and works on every rank (Sec. 3.2).

5.5 Geodesic orbits on an AMR mesh

The grid-mode samplers read the metric from the refinement level that contains the particle, which makes them usable on Carpet's AMR meshes without modification. We verify this on the same circular orbit as the unigrid reference `conv100.par` (Kerr–Schild, $M = 2$, $a = 0.5$, particle on the $r = 10$ equatorial circle, integrated to $t = 150.1$, ~ 1.05 orbital periods, $T \simeq 143$), now evolved on a 3-level AMR mesh: a 50^3 base grid ($\Delta x = 1$) with three refinement levels (factor 2) of a box centred on the origin (half-width 16 cells on the first level, then 8; the innermost box is refined twice, so the finest level has $\Delta x = 1/8$). The orbiting particle lies on the first refinement level ($\Delta x = 1/2$) for most of its path — the same local resolution as the 100^3 unigrid reference — and crosses onto the second level ($\Delta x = 1/4$) near the box diagonals; the two inner levels resolve the central region. The particle integrator is otherwise unchanged (RKF45, same seed and initial data as the reference run), so the only differences between the two set-ups are the grid structure and the driver's 4:1 subcycling.

Table 3: AMR (3 levels) vs. unigrid reference (conv100): maxima over the 16 dump times $t = 0.1, 10.1, \ldots, 150.1$ of the Cartesian position difference $\max_i |\Delta x^i|$, the radial difference $|\Delta r|$, and the orbital phase difference $|\Delta\phi|$.

sampler	$\max_i \lvert\Delta x^i\rvert$	$\max \lvert\Delta r\rvert$	$\max \lvert\Delta\phi\rvert$ (rad)
local27 (default)	1.2×10^{-1}	4.3×10^{-2}	1.3×10^{-2}
aei (order 4)	1.4×10^{-1}	5.0×10^{-2}	1.6×10^{-2}
carpet (order 2)	3.4×10^{-1}	2.3×10^{-1}	4.2×10^{-2}

All three samplers keep the particle on a stable orbit throughout the integration (no blow-up, no lost particle), with the deviation from the unigrid reference bounded at the level expected from the sampler's formal order: the two higher-fidelity samplers stay within 5×10^{-2} (0.5%) of the reference in radius and 1.7×10^{-2} rad ($\sim 1°$) in phase, while the second-order `carpet` sampler drifts to 2.3×10^{-1} (2.3%) in radius — its pointwise composition of the Christoffel symbols from interpolated first derivatives amplifies the second-order interpolation error, in the same range as the $\max |\Delta L_z| \sim 4 \times 10^{-2}$ drifts of the original `CarpetInterp`-based implementation on the precessing ellipse of Sec. 5.4. At the first dump after the mesh settles ($t = 10.1$) all three deviations are already at their small-t level ($\lesssim 5 \times 10^{-3}$ in position), so no transient is associated with the refinement. The pointwise samplers therefore resolve the metric correctly on an AMR mesh, including across the level boundaries, and the geodesic equation can be integrated in a Carpet AMR evolution (with its 4:1 subcycling) without any modification of the integrator.

5.6 Robustness on a non-stationary merger background

The checks above all use a static Kerr–Schild field, where the dominant error source is the grid's own discretization. A production use case, however, is a spacetime whose metric oscillates with time

at every grid point. To check that the four grid samplers remain well posed there, we integrated 100 test particles on the `GW150914` binary black hole merger (the first directly detected gravitational-wave signal [LIGO Scientific Collaboration and Virgo Collaboration, 2016]; LIGO masses 36.2 and $29.6\,M_\odot$), a Two–Punctures BSSN evolution of the spinning pair (total mass $M = 1$) on a moving-box AMR mesh that tracks both punctures, over 2048 Cactus steps. There is no analytic reference orbit in a non-stationary field, so the relevant checks are stability and mutual agreement, not absolute accuracy: none of the four samplers (`aei`, `carpet`, `local27`, `local27lin`) produced a divergent or `NaN` state, no particle was lost, and the four runs agreed pairwise to within $\sim 8 \times 10^{-3}$ in particle position over the whole run. This is at the level of the coherent grid discretization error that dominates the static benchmarks of Sec. 5.4, and it confirms that the pointwise and default samplers are robust on a genuinely dynamical background, not merely on a frozen field.

6 Strong scaling

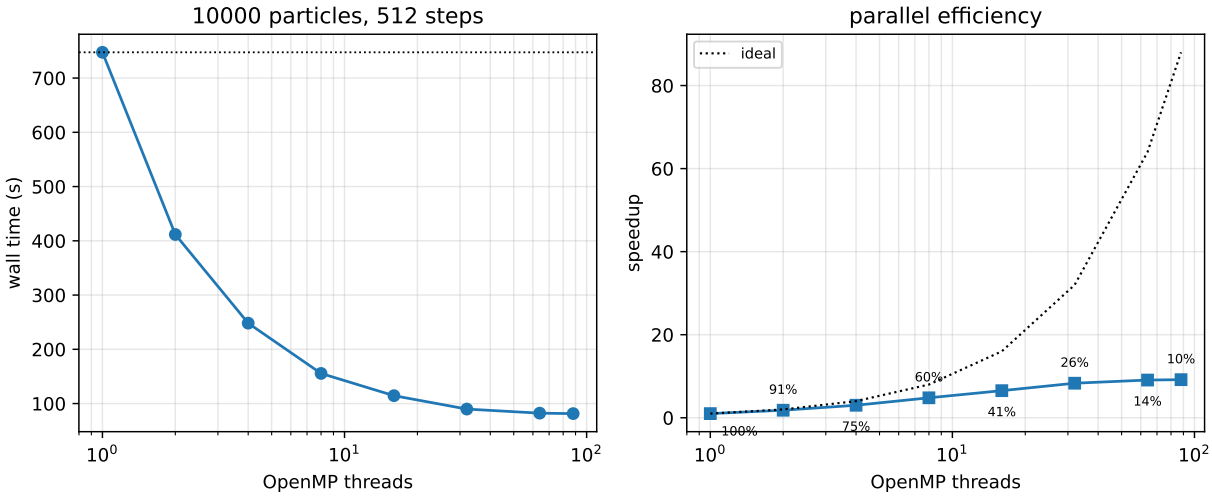


Figure 6: Strong scaling with 10^4 particles, 50^3 grid, 512 steps, `local27` sampler. Left: wall time versus number of OpenMP threads. Right: speedup versus thread count (solid line, ideal) with the parallel efficiency annotated at each point.

The per-particle independence of the integration loop (Sec. 4) should translate directly into parallel speedup. We run a configuration with 10^4 particles (the three orbits of Table 1 plus 9997 particles with random positions inside the safe box and zero velocity, fixed random seed), 50^3 grid, `Exact = no`, 512 steps, on a two-socket node with 44 physical cores (88 logical threads via hyperthreading), varying only `OMP_NUM_THREADS` from 1 to 88. Figure 6 shows the wall time and the parallel efficiency. The speedup grows with the thread count up to 32 threads (8.3×) and then plateaus at 9.2× (efficiency 10.4%) at 88 threads. A per-phase timing breakdown (Cactus `TimerReport`) attributes the plateau to two effects. First, the surrounding Cactus framework (metric bookkeeping, boundary conditions, NaN checks) consumes ∼40 ms per step, nearly independent of the thread count (19.0 s total at one thread versus 22.9 s at 88): about 3% of the ∼1.45 s single-thread step but about 29% of the ∼0.15 s 88-thread step. Second, the parallel particle region itself scales sublinearly: it shrinks from ∼1.41 s to ∼0.104 s per step, a speedup of only 13.7× on 88 threads, consistent with the 44

physical (hyperthreaded) cores, cross-NUMA stencil reads, and OpenMP scheduling overhead.

As a complementary measurement in a production setting, we ran the tracker inside full `GRHydro` evolutions [Font, 2008] with 5000 and with 10^4 particles; each run shared the 88-thread node with three other concurrent instances, i.e. only ~ 22 threads were available per instance. The particle integration (`geodesics_integrate` in the `TimerReport`) consumed 49.7 s of 2860.5 s total wall time (1.7%) in the 5000-particle run and 84.1 s of 2749.1 s (3.1%) in the 10^4-particle run: the dominant costs remain the hydrodynamic reconstruction ($\sim 27\%$) and the Riemann solves ($\sim 10\%$), and doubling the particle count grows the integration cost by a factor of only 1.7. Because these fractions were measured under contention from the co-running instances, the overhead in a dedicated run is smaller still: tracking test particles alongside an evolution is essentially free compared to the evolution itself.

7 Conclusion and availability

We have presented `Geodesic`, an Einstein Toolkit thorn that integrates test-particle geodesics in Kerr–Schild Cartesian coordinates. The thorn supports an exact analytic Kerr–Schild metric (grid-independent, verified to be a vacuum solution at machine precision) and a grid mode with three pointwise samplers selected by one parameter — the default 27-point local Lagrange interpolant, a fourth-order pointwise interpolant built on `AEILocalInterp`, and a second-order pointwise interpolator (historically named after the `CarpetInterp` driver) — with an optional linear-in-time blend of the two most recent grid levels. The grid mode assumes no particular analytic form of the metric; for stationary grids $\partial_t g_{\mu\nu} = 0$ is detected automatically, and for time-dependent grids the time derivative is recovered from the three most recent `ADMBase` time levels (Sec. 3): with `Exact = no`, the thorn tracks geodesics in whatever spacetime the grid carries, so it can be used with any gridded metric, for example the spacetime of a full numerical-relativity evolution on a fixed background. In that mode each particle is an independent unit of work, and an OpenMP parallelization of the particle loop scales to 10^4 particles, reaching a speedup of $9.2\times$ (10.4% efficiency) on the 88 logical threads of a two-socket, 44-core hyperthreaded node. Verification against an independent reference implementation shows conservation of E, L_z and the norm to $\sim 10^{-9}$ over $10^3\,M$, a measured local convergence order of 3.7 (formal order three) of the default sampler, and, on a precessing orbit of 100 particles close to the plunge boundary, a drift ordering of `aei` < `local27` < `carpet` (Sec. 5.4); and, on a 3-level AMR mesh, orbits of the three grid samplers that reproduce the unigrid reference to within 10^{-2} (Sec. 5.5).

The thorn is released under the GPLv2+ license together with the configuration files used in this work; the code, the configuration files and this paper are publicly available at `https://github.com/wylyhzh/Geodesic`.

Statement on AI assistance

During the preparation of this paper the author used AI assistants (large language models) for debugging the code, drafting and revising the text, and producing the figures. All simulations, data and results are the author's own; every number in the paper was checked by the author, who takes full responsibility for its content.

References

James M. Bardeen. Timelike and Null Geodesics in the Kerr Metric. In Charles DeWitt and Brian S. DeWitt, editors, *Black Holes (Les Astres Occlus)*, pages 215–239. Gordon and Breach, New York, 1973.

James M. Bardeen, William H. Press, and Saul A. Teukolsky. Rotating black holes: Locally nonrotating frames, energy extraction, and scalar synchrotron radiation. *The Astrophysical Journal*, 178:347–370, 1972. doi: 10.1086/151796.

Jean-Paul Berrut and Lloyd N. Trefethen. Barycentric Lagrange Interpolation. *SIAM Review*, 46 (3):501–517, 2004. doi: 10.1137/s0036144502417715.

R. D. Blandford and R. L. Znajek. Electromagnetic extraction of energy from Kerr black holes. *Monthly Notices of the Royal Astronomical Society*, 179(3):433–456, 1977. doi: 10.1093/mnras/179.3.433.

Cactus Code. The Cactus computational framework. http://www.cactuscode.org/, 2024. Accessed: 2026-09-06.

Einstein Toolkit. AEILocalInterp thorn documentation, Einstein Toolkit thorn guide. https://einsteintoolkit.org/thornguide/Numerical/AEILocalInterp/documentation.html, 2026a. Accessed: 2026-09-15.

Einstein Toolkit. CarpetInterp thorn documentation, Einstein Toolkit thorn guide. https://einsteintoolkit.org/thornguide/Carpet/CarpetInterp/documentation.html, 2026b. Accessed: 2026-09-15.

E. Fehlberg. Klassische Runge-Kutta-Formeln fünfter und siebenter Ordnung mit Schrittweiten-Kontrolle. *Computing*, 4(2):93–106, 1969. doi: 10.1007/BF02234758.

José A. Font. Numerical Hydrodynamics and Magnetohydrodynamics in General Relativity. *Living Reviews in Relativity*, 11:7, 2008. doi: 10.12942/lrr-2008-7.

Samuel E. Gralla and Alexandru Lupsasca. Null geodesics of the Kerr exterior. *Physical Review D*, 101(4):044032, 2020. doi: 10.1103/PhysRevD.101.044032.

R. P. Kerr. Gravitational field of a spinning mass as an example of algebraically special metrics. *Physical Review Letters*, 11(23):237–238, 1963. doi: 10.1103/PhysRevLett.11.237.

R. P. Kerr and A. Schild. Some Algebraically Degenerate Solutions of Einstein's Gravitational Field Equations. *Proceedings of Symposia in Applied Mathematics*, 17:199–209, 1965. doi: 10.1090/psapm/017/0216846.

LIGO Scientific Collaboration and Virgo Collaboration. GW150914: Observation of gravitational waves from a binary black hole merger. *Physical Review Letters*, 116(6):061102, 2016. doi: 10.1103/PhysRevLett.116.061102. URL https://arxiv.org/abs/1602.03837.

F. Löffler, J. Faber, E. Bentivegna, T. Bode, P. Diener, R. Haas, I. Hinder, B. C. Mundim, C. D. Ott, E. Schnetter, et al. The Einstein Toolkit: a community computational infrastructure for relativistic astrophysics. *Classical and Quantum Gravity*, 29(11):115001, 2012. doi: 10.1088/0264-9381/29/11/115001.

Claudia Moreno, Darío Núñez, and Olivier Sarbach. Kerr–Schild-type initial data for black holes with angular momenta. *Classical and Quantum Gravity*, 19(23):6059–6073, 2002. doi: 10.1088/0264-9381/19/23/312.

Eric Poisson. The Motion of Point Particles in Curved Spacetime. *Living Reviews in Relativity*, 7: 6, 2004. doi: 10.12942/lrr-2004-6.

D. C. Wilkins. Bound geodesics in the Kerr metric. *Physical Review D*, 5(4):814–822, 1972. doi: 10.1103/PhysRevD.5.814.